DIC17
Latches and Flip-Flops

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Compare static, dynamic, and weak-feedback DFFs following handout PDF pages 74–77, covering hazards, noise immunity, frequency limits, and asynchronous reset. The circuits use transistors and TGs rather than ideal edge-triggered behavioral models.

2 Handout requirements and test mapping

Requirement	Deck prefix and conditions	Status
Step 1: hazards, latch versus DFF	01; 20 ns period	Waveforms measured
Step 1: operating-frequency limit	03; 12 periods	Finite boundary located
Step 2: equal-noise comparison	02; 1/5/20 fF	Measured
Fig. 17.1(c): weak feedback	01/02	Compared
Q1–3: overlap, slope, setup/hold	Theory; Lab21 sweep	Analyzed; not full extraction
Q4: reset-low DFF	04; both phases	Measured

3 Circuit and timing derivation

Two TG latches use complementary clocks: the master is transparent during low phase and the slave during high phase, producing a positive-edge capture path. The static circuit uses switched feedback at both storage nodes. The dynamic circuit removes feedback and stores charge parasitically. The weak-feedback circuit uses long-channel weak inverters. Feedback loading, restoration current, and contention with new data jointly affect speed.

Asynchronous reset replaces the slave output inverter with a NOR: $Q = \overline{Z + R}$, forcing low when $R = 1$. This verifies output reset, not a complete reset recovery/removal timing signoff.

4 Simulation setup and interpretation

The supplied encrypted MOS model is evaluated at a baseline supply of 1.0 V and 25 °C. The 1.0 V value is a teaching condition, not a verified voltage rating. These results do not reproduce the handout's original process or supply conditions. Unless stated in the deck, $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m, bulks are tied to their corresponding rails, and output load is 5 fF. Transmission-gate and weak-feedback dimensions are specified explicitly in each deck.

PrimeSim HSPICE S-2021.09 win64 was used with `RUNLVL=6`, `ACCURATE`, and `WARNLIMIT=100`. Every baseline and `.alter` analysis concluded without errors or failed measurements. The simulator still reports suppressed warnings associated with encrypted blocks; this does not establish that the model is warning-free or that those warnings are harmless. The `.tran` print interval is not a fixed integration step: the solver uses adaptive timesteps. A finite sweep is not a complete process qualification.

Unless stated otherwise, the declared functional diagnostic uses $V \leq 0.2$ V for low and $V \geq 0.8$ V for high. These are report-level interpretation thresholds, not numerical handout requirements or extracted receiver noise margins. Correct samples do not exclude glitches between samples. Figures are redrawn from actual simulation data; `results/` contains per-case waveforms, measurements, and derived comparison CSV files.

Public decks omit model loading paths; model configuration is added only to local execution copies. The model itself is not distributed. The first two filename digits identify the deck numbers used below. Every deck is standalone.

5 Results

The three circuits produce approximately 1, 0, 0, and 0 V at 7, 27, 47, and 67 ns. Pulses during the closed phase do not directly propagate to the output. The 45 ns data transition deliberately lies near the sampling edge and the setup/hold boundary; one waveform cannot prove that all near-edge hazards are safely rejected.

Coupling (fF)	Static peak (V)	Dynamic peak (V)	Weak-feedback peak (V)
1	0.00263	0.01208	0.01641
5	0.00375	0.00814	0.01172
20	0.06041	1.04148	0.99972

Under the strong 20 fF disturbance, the static output peaks at only 0.06041 V, while dynamic and weak-feedback outputs reach approximately 1.04148 and 0.99972 V, producing erroneous highs. Weak feedback does not reject arbitrarily strong disturbances. The small 1/5 fF peaks need not be monotonic because the measurement includes different injection phases and restoration dynamics.

5.1 Frequency boundary and reset

Period (ns)	Frequency (GHz)	Static correct/8	Dynamic correct/8
2	0.5000	8	8
1.8	0.5556	8	8
1.6	0.6250	8	8
1.4	0.7143	4	8
1.2	0.8333	4	8
1	1.0000	0	4

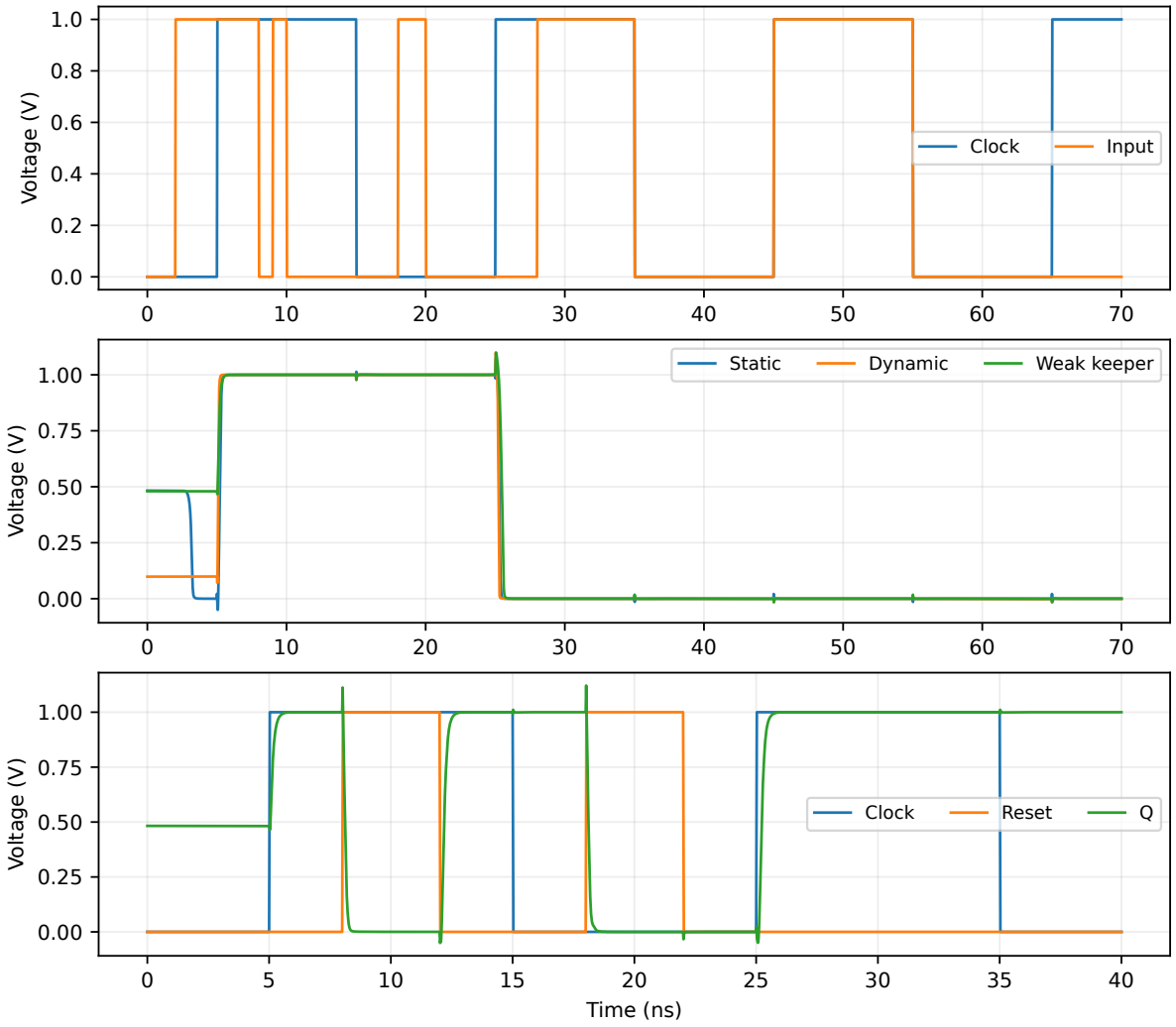


Figure 1: The upper plots show data hazards and the three DFF outputs; the lower plot shows asynchronous reset in both clock phases.

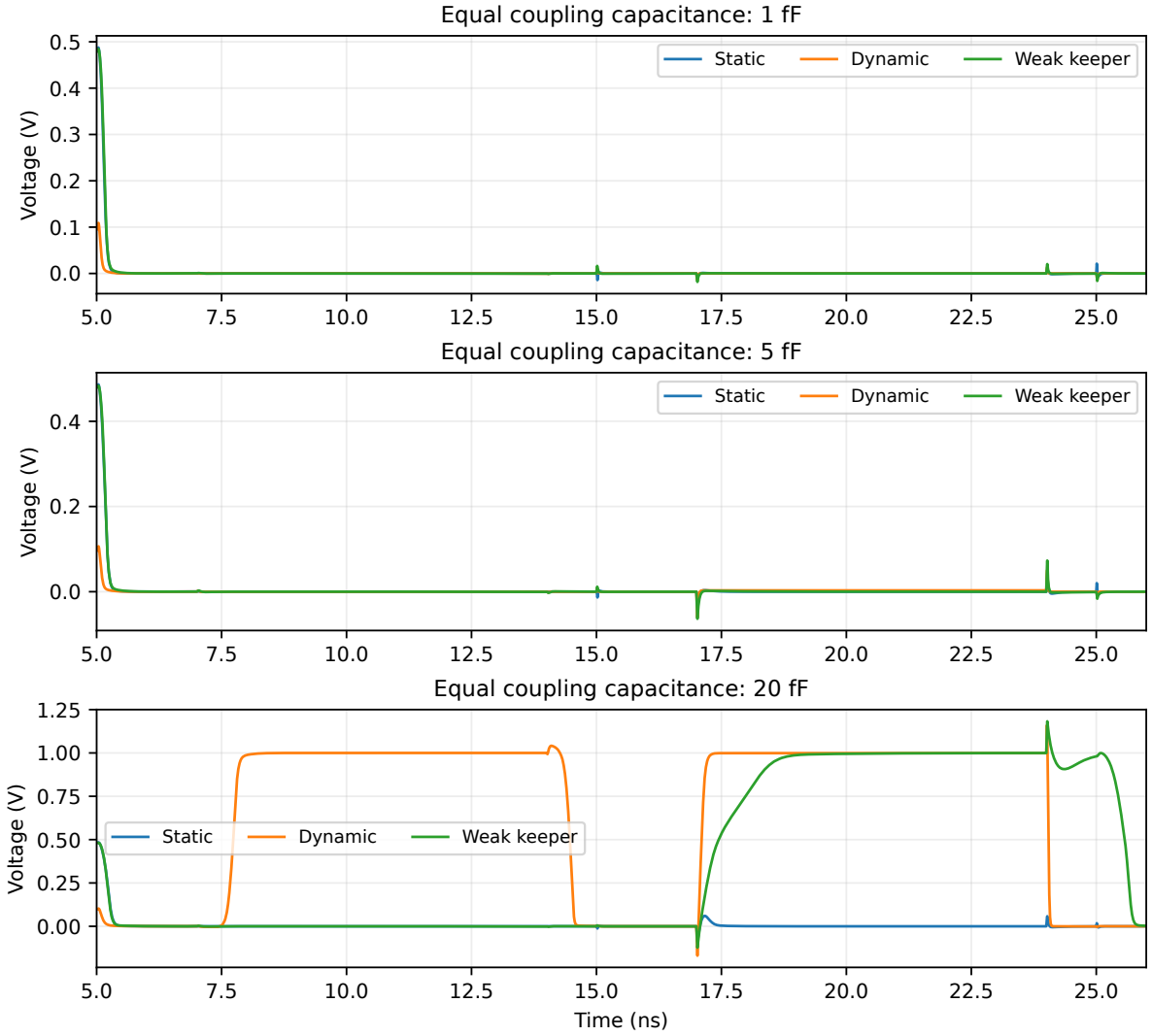


Figure 2: Identical capacitors and aggressors couple into the master/slave storage nodes with input fixed at 0.

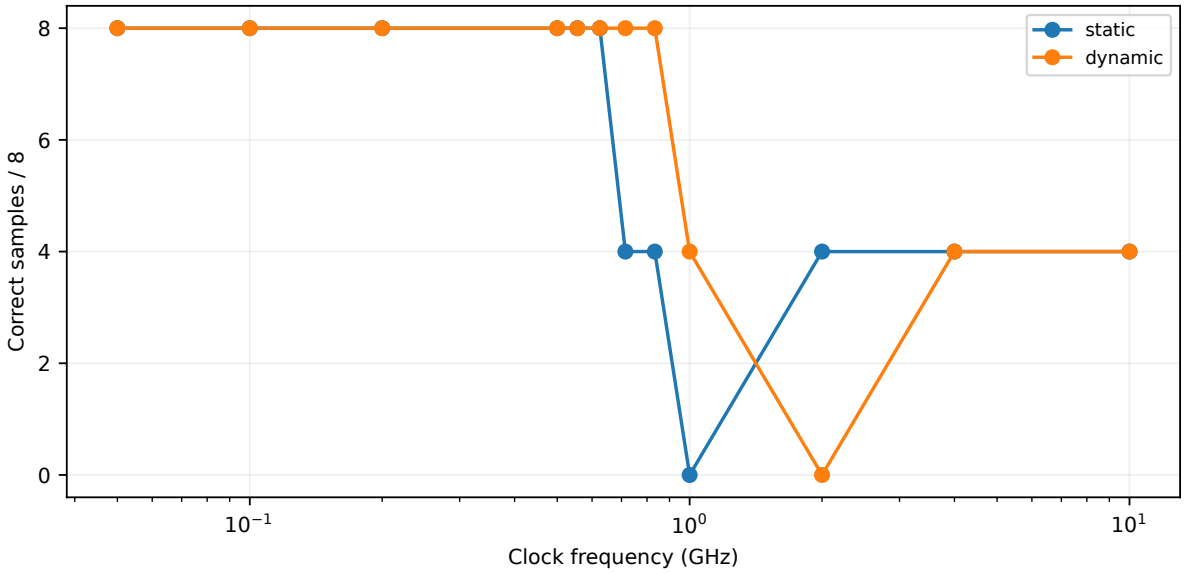


Figure 3: Fixed-time interpretation for 12 tested periods; connecting lines guide the eye and do not verify continuous frequencies.

The clock begins rising at $0.25T$ and outputs are sampled at $0.49T$, allowing about $0.24T$ to settle. The highest tested frequencies with 8/8 correct samples are 0.625 GHz (1.6 ns) for static and 0.8333 GHz (1.2 ns) for dynamic. Adjacent failing points are 0.7143 GHz (1.4 ns) and 1 GHz, respectively. These are finite-sweep boundaries under the stated load, 20 ps clock edge, and observation deadline, not unconditional $f_{\max}$ values.

Output during reset in high and low clock phases is 0.0001976 V and 3.395 μV, respectively; recapture after release gives 0.999979 V.

6 Discussion, differences, and unverified scope

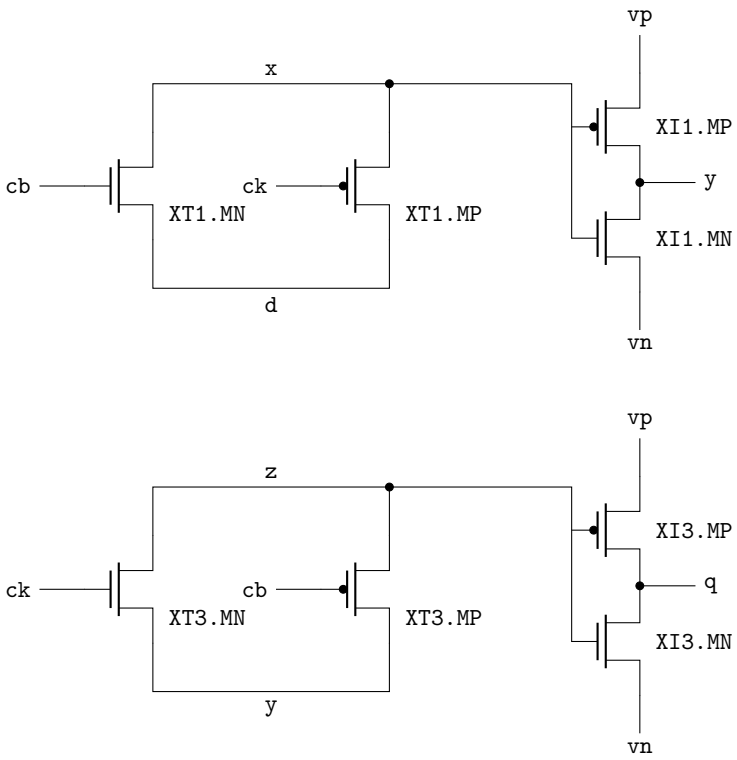
Q1: overlapping conduction of complementary clocks can make master and slave transparent simultaneously and allow race-through. Q2: slow edges lengthen partial conduction and contention and alter available settling time, so nominal period alone is insufficient. Q3: setup allows the master storage node to settle before closing; hold prevents new data from crossing before closure completes. Both depend on clock slope, TG delay, and regeneration, rather than zero-time ideal logic. Continuous setup/hold and metastability extraction is not claimed; Lab21 separately measures slew and skew. Equal output loads and equal injected disturbances are used to avoid attributing mismatched tests to noise immunity.

7 MOS transistor schematic

Source netlist: `01_hazards.sp`. Dynamic, static, and weak-keeper D flip-flops in the hazard comparison.

Identical node labels denote electrical connections. The drawings are editable CircuitikZ source; hierarchical boxes below refer to the MOS cells shown here.

Unless explicitly labeled otherwise, every NMOS body connects to `vn`=0 and every PMOS body to `vp`=V_{DD}; body wires are omitted for readability. Ordinary inverter and dynamic/clocked cells use $W_n/L_n = 0.6/0.25\,\mu$m, $W_p/L_p = 1.8/0.25\,\mu$m. Both TG devices use $W/L = 0.6/0.25\,\mu$m.



Input `d` enters XT1; XI3 drives `q`. For the dynamic `dff`, this is the complete 8-transistor signal path.

Variant	Added feedback connections
`sdff` (16T)	XI2: INV(`y`→`fb1`); XT2: TG(`fb1`↔`x`, en=ck, eb=cb). XI4: INV(`q`→`fb2`); XT4: TG(`fb2`↔`z`, en=cb, eb=ck).
`kdff` (12T)	XK1: weak INV(`y`→`x`); XK2: weak INV(`q`→`z`).

The weak cell has the same two-MOS inverter topology, with PMOS $W/L = 0.3/1\,\mu$m and NMOS $W/L = 0.3/3\,\mu$m. Top instances `Xd`, `Xs`, and `Xk` drive `qd`, `qs`, and `qk`, respectively. Each output has 5 fF to 0. All share input `d`, clock `clk`, and `cb`=$V_{DD} - V($`clk`$)$, with $V_{DD} = 1$ V. The TG labels give both actual control pins, so the feedback phases remain explicit.